

DETAILED BILL OF QUANTITIES — CONSTRUCTION COST ESTIMATE

Project: Proposed Multi-Storey Residential Complex, Sector 14, Gurugram
Estimate Ref: CCE-2024-GGN-047 | Prepared: April 2024 | Standard: IS:1200 Method of Measurement

PART A — SUBSTRUCTURE (BELOW PLINTH)

A1. EARTH WORK AND EXCAVATION

Item No.	Description	Unit	Qty	Rate (INR)	Amount (INR)	Remarks
A1.01	Excavation in hard soil for foundations by mechanical means	m3	2850.0	185.00	527,250	IS:1200-I
A1.02	Disposal of excavated earth within 1km lead	m3	2850.0	95.00	270,750	Tipper rate
A1.03	Anti-termite treatment to soil below plinth	m2	1200.0	42.00	50,400	IS:6313
A1.04	Sand filling in plinth area, compacted	m3	380.0	1,250.00	475,000	Compacted 95% MDD
A1.05	Lean concrete PCC M10 below foundations, 100mm	m3	142.0	4,800.00	681,600	

A2. FOUNDATION CONCRETE — ISOLATED FOOTINGS

Item No.	Footing ID	Size (mm)	Depth (mm)	Conc. Grade	Qty (m3)	Rate (INR/m3)	Amount (INR)
A2.01	FTG-A01 to A06 (6 nos)	2500x2500	600	M30	22.50	5,800	130,500
A2.02	FTG-B01 to B08 (8 nos)	2800x2800	700	M30	43.90	5,800	254,620
A2.03	FTG-C01, C02 (2 nos)	3200x3200	750	M25	15.36	5,200	79,872
A2.04	FTG-D01 to D04 (4 nos) — High 4000x4000	4000x4000	900	M30	57.60	5,800	334,080
A2.05	Pile Caps PC-01 to PC-06 (6 nos)	3000x2000	900	M25	21.60	5,200	112,320
A2.06	Raft Foundation — Core Wall Zone	10000x12000	1000	M30	216.00	5,800	1,252,800
					Sub-total A2		2,164,192

INCONSISTENCY: Items A2.03 (FTG-C01/C02) and A2.05 (Pile Caps) specify M25 concrete. material_specifications.csv (MAT-075) specifies M30 for pile caps and site_investigation_report.pdf Section 8 specifies M30 for isolated footings. Correct grade is M30 for all footings and pile caps. Corrected amounts: A2.03 = 15.36 x 5800 = INR 89,088; A2.05 = 21.60 x 5800 = INR 125,280.

A3. FOUNDATION REINFORCEMENT STEEL

Item No.	Description	Unit	Qty (MT)	Rate (INR/MT)	Amount (INR)	Remark
A3.01	Fe500 HYSD bars in isolated footings (all dia)	MT	28.50	58,500	1,667,250	IS:1786
A3.02	Fe500 HYSD bars in pile caps	MT	8.20	58,500	479,700	IS:1786
A3.03	Fe500 HYSD bars in raft foundation	MT	42.80	58,500	2,503,800	IS:1786
A3.04	Fe500 HYSD bars in tie beams	MT	12.40	58,500	725,400	IS:1786
					Sub-total A3	5,376,150

A4. BORED CAST-IN-SITU PILES

Item No.	Description	Unit	Qty	Rate (INR)	Amount (INR)	Remark
A4.01	600mm dia bored pile, L=15m, M30 concrete with full-length cages	m	24	185,000	4,440,000	IS:2911
A4.02	Pile integrity test (Low Strain PIT)	nos	24	4,500	108,000	100% testing
A4.03	Pile load test (Initial)	nos	2	280,000	560,000	1.5x design load
					Sub-total A4	5,108,000

PART A TOTAL — SUBSTRUCTURE

Section	Description	Amount (INR)
A1	Earthwork and Excavation	2,005,000
A2	Foundation Concrete	2,164,192
A3	Foundation Reinforcement Steel	5,376,150
A4	Bored Piles	5,108,000
	PART A TOTAL	14,653,342

PART B — SUPERSTRUCTURE (ABOVE PLINTH TO ROOF)

B1. STRUCTURAL CONCRETE — COLUMNS

Item No.	Description	Unit	Qty (m3)	Rate (INR/m3)	Amount (INR)	Remark
B1.01	Columns GF to 3F, 400x400mm, M30 concrete	m3	88.60	5,800	513,880	IS:456
B1.02	Columns 4F to Roof, 350x350mm, M25 concrete	m3	62.40	5,200	324,480	IS:456
B1.03	Corner cols 500x500mm GF to 3F, M30 concrete	m3	18.00	5,800	104,400	IS:456
B1.04	High-axial cols 450x450mm all floors, M30 concrete	m3	32.40	5,800	187,920	IS:456
				Sub-total B1	1,130,680	

B2. STRUCTURAL CONCRETE — BEAMS

Item No.	Description	Unit	Qty (m3)	Rate (INR/m3)	Amount (INR)	Remark
B2.01	Primary beams 300x600mm typical, M25 concrete	m3	198.40	5,200	1,031,680	IS:456
B2.02	Secondary beams 250x550mm, M25 concrete	m3	64.20	5,200	333,840	IS:456
B2.03	Transfer beams 450x800mm MEM-048/049, M35 concrete	m3	14.40	6,500	93,600	IS:456
B2.04	Transfer beams 400x750mm MEM-014/015/116, M35 concrete	m3	24.30	6,500	157,950	IS:456
B2.05	Podium beams 375-425mm wide, M30 concrete	m3	38.60	5,800	223,880	IS:456
B2.06	Roof beams 300x600mm, M25 concrete	m3	42.80	5,200	222,560	IS:456
				Sub-total B2	2,063,510	

B3. STRUCTURAL CONCRETE — SLABS

Item No.	Description	Unit	Qty (m3)	Rate (INR/m3)	Amount (INR)	Remark
B3.01	Two-way slabs t=150mm, M25 concrete (all floors)	m3	648.00	5,200	3,369,600	IS:456
B3.02	Slabs t=160mm (heavy load zones), M25 concrete	m3	96.00	5,200	499,200	IS:456
B3.03	Terrace slab t=140mm, M25 concrete	m3	72.00	5,200	374,400	IS:456
B3.04	Staircase waist slab t=150mm, M20 concrete	m3	18.60	4,600	85,560	INCONSISTENCY B3.04: WRONG GRADE per MAT-036
				Sub-total B3	4,328,760	

B4. REINFORCEMENT STEEL — SUPERSTRUCTURE

Item No.	Description	Unit	Qty (MT)	Rate (INR/MT)	Amount (INR)	Remark
B4.01	Fe500 HYSD in columns (all sizes, all floors)	MT	124.80	58,500	7,300,800	IS:1786
B4.02	Fe500 HYSD in primary beams (B2.01, B2.06)	MT	198.60	58,500	11,618,100	IS:1786
B4.03	Fe500 HYSD in transfer beams (B2.03, B2.04)	MT	62.40	58,500	3,650,400	IS:1786
B4.04	Fe500 HYSD in two-way slabs (B3.01 to B3.03)	MT	98.40	55,000	5,406,000	INCONSISTENCY B4.04: Fe500 HYSD uses Fe415 price (55,000) but spec is Fe500
B4.05	Fe500D HYSD in shear walls (ductile detailing)	MT	42.80	60,000	2,568,000	IS:13920
B4.06	Fe500D HYSD in core walls and lift pit	MT	28.60	60,000	1,716,000	IS:13920
			Sub-total B4		32,265,300	

INCONSISTENCY B4.04: Fe500 HYSD steel in slabs is priced at INR 55,000/MT (Fe415 rate). Correct rate per material_specifications.csv (MAT-001) is INR 58,500/MT. Correct amount = 98.40 x 58,500 = INR 5,756,400 (difference = INR 344,400 under-estimated).

PART B TOTAL — SUPERSTRUCTURE

Section	Description	Amount (INR)
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B1	Columns Concrete	1,130,680
B2	Beams Concrete	2,063,510
B3	Slabs Concrete	4,328,760
B4	Reinforcement Steel Superstructure	32,265,300
	PART B TOTAL	39,788,250

PART C — FINISHING AND EXTERNAL DEVELOPMENT

Item No.	Description	Unit	Qty	Rate (INR)	Amount (INR)	Remark
C1.01	External plaster CM1:4, 20mm thick	m2	4200	285	1,197,000	IS:1661
C1.02	Internal plaster CM1:6, 12mm thick	m2	12600	165	2,079,000	IS:1661
C1.03	AAC block 600 partition walls, 200mm thick	m2	2800	820	2,296,000	IS:2185
C1.04	External clay brick 230mm wall CM1:6	m2	1800	1,250	2,250,000	IS:2212
C1.05	Ceramic floor tiles 600x600, 10mm thick	m2	8400	680	5,712,000	IS:13630
C1.06	Vitrified floor tiles 800x800, common areas	m2	1200	1,450	1,740,000	IS:13630
C1.07	Terrace waterproofing APP bituminous membrane	m2	1200	620	744,000	IS:1322
C1.08	Painting external face two coats	m2	4200	145	609,000	IS:101
C1.09	Painting internal walls two coats	m2	12600	95	1,197,000	IS:101
C1.10	Aluminium glazed windows, DGU	m2	840	4,800	4,032,000	IS:1948
C1.11	Solid core flush doors 35mm thick	nos	280	8,500	2,380,000	IS:2202
C1.12	Mild steel stair railings, 38mm sq bar	rm	420	1,850	777,000	IS:2062
C1.13	Compound wall and boundary treatment	rm	280	4,200	1,176,000	Perimeter
C1.14	Paving block in external areas 80mm	m2	1600	380	608,000	IS:15658
C1.15	Landscaping and soft areas	LS	1	850,000	850,000	Lump sum
				Sub-total C		27,647,000

PART D — MEP SERVICES

Item No.	Description	Unit	Qty	Rate (INR)	Amount (INR)	Remark
D1.01	Water supply GI pipes complete	LS	1	2,800,000	2,800,000	IS:1239
D1.02	Sanitary drainage UPVC pipes complete	LS	1	1,950,000	1,950,000	IS:4985
D1.03	Electrical wiring and DB boards	LS	1	4,200,000	4,200,000	IS:732
D1.04	Fire fighting system sprinkler and hydrant	LS	1	2,100,000	2,100,000	NBC 2016
D1.05	Lifts 2 nos passenger 8-person	nos	2	1,800,000	3,600,000	IS:14665
D1.06	Air conditioning VRF system	TR	120	28,500	3,420,000	IS:1391
D1.07	STP sewage treatment plant 50 KLD	LS	1	1,850,000	1,850,000	MOEF norms
				Sub-total D	19,920,000	

GRAND SUMMARY — COMPLETE PROJECT

Part	Description	Amount (INR)	Verified
A	Substructure	14,653,342	See Note 1
B	Superstructure	39,788,250	See Note 2
C	Finishing and External Development	27,647,000	OK
D	MEP Services	19,920,000	OK
	Sub-Total (A+B+C+D)	101,908,592	INCONSISTENCY: Arithmetic wrong here
	Contingency @ 5%	5,095,430	Calculated on wrong sub-total
	GRAND TOTAL (incl. contingency)	107,003,022	WRONG — see note below

INCONSISTENCY NOTE — GRAND TOTAL ARITHMETIC ERROR: Sub-Total should be 14,653,342 + 39,788,250 + 27,647,000 + 19,920,000 = 102,008,592. The document records 101,908,592 (difference = INR 100,000 — a transcription error in Part C)

sub-total). Correct 5% contingency = $102,008,592 \times 0.05 = 5,100,430$. Correct Grand Total = $102,008,592 + 5,100,430 = 107,109,022$. Document shows 107,003,022 — incorrect by INR 106,000.

Prepared by: Quantity Surveyor | Checked by: Senior QS | Approved by: Project Director | All rates inclusive of labour, material and overheads. GST extra as applicable.